

# Aggregation and Endogeneity (BLP)

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## **First: Some Review**

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## What is the goal?

Recall the multi-product Bertrand FOCs from Week 2, where firms solve  $\arg \max_{p \in \mathcal{J}_f} \pi_f(\mathbf{p}) = \sum_{j \in \mathcal{J}_f} (p_j - c_j) \cdot q_j(\mathbf{p})$ , which rearrange to:

$$p_j(p_{-j}) = \underbrace{\frac{1}{1 + 1/\epsilon_{jj}(\mathbf{p})}}_{\text{Markup}} \left[ c_j + \sum_{k \in \mathcal{J}_f \setminus j} (p_k - c_k) \cdot D_{jk}(\mathbf{p}) \right].$$

We call  $D_{jk}(\mathbf{p}) = \frac{\frac{\partial q_k}{\partial p_j}(\mathbf{p})}{\left| \frac{\partial q_j}{\partial p_j}(\mathbf{p}) \right|}$  the **diversion ratio** and  $\epsilon_{jj}$  the **own elasticity** and these are the main deliverables.

## Starting Point: From Individual Choice to Market Shares

Last lecture: individual choices  $(d_{i1}, \dots, d_{iJ}, d_{i0}) \sim \text{Categorical}(\sigma_{i1}, \dots, \sigma_{iJ}, \sigma_{i0})$  with mixed logit choice probabilities:

$$\sigma_{ij}(\theta) = \mathbb{P}(d_{ij} = 1) = \int \frac{\exp[\beta_i x_j]}{1 + \sum_{k \in \mathcal{J}} \exp[\beta_i x_k]} f(\beta_i | \theta) \partial \beta_i$$

If each individual is **exchangeable** then ex-ante they have the same choice probabilities:  $\sigma_{ij} = \sigma_j$ , and the sum of  $M$  Categoricals is Multinomial:

$$(q_1^*, \dots, q_J^*, q_0^*) \sim \text{Mult}(M, \sigma_1, \dots, \sigma_J, \sigma_0)$$

where  $q_j^* = \sum_{i=1}^M d_{ij}$  is a **sufficient statistic**.

- ▶ If  $M$  gets large enough then  $(\frac{q_1}{M}, \dots, \frac{q_J}{M}, \frac{q_0}{M}) \rightarrow (s_1, \dots, s_J, s_0)$
- ▶ Idea: Equate observed market shares to the conditional choice probabilities  $(\sigma_1(\mathbf{x}_i, \theta), \dots, \sigma_J(\mathbf{x}_i, \theta), \sigma_0(\mathbf{x}_i, \theta))$ .
- ▶ Challenges: We probably don't really observe  $q_0$  and hence  $M$ .
- ▶ Introduce idea of market  $t$  (otherwise not much data!)
- ▶ MLE on individual data (and why MSLE is **inconsistent** under simulation error) was covered in the mixed logit lecture.

# Lots of papers stop here

Table IV. Hospital demand results, ML estimation

| Interaction Terms              | Variable               | Estimated coefficient |
|--------------------------------|------------------------|-----------------------|
| Interactions: Teaching         | Distance (miles)       | -0.215** (0.004)      |
|                                | Distance squared       | 0.001** (0.000)       |
|                                | Emergency* distance    | -0.008** (0.004)      |
|                                | Cardiac                | 0.090 (0.060)         |
|                                | Cancer                 | 0.192** (0.069)       |
|                                | Neurological           | 0.546** (0.175)       |
|                                | Digestive              | -0.145** (0.062)      |
|                                | Labor                  | 0.157** (0.048)       |
|                                | Newborn baby           | 0.038 (0.075)         |
|                                | Income (\$000)         | 0.007** (0.001)       |
| Interactions: Nurses per bed   | PPO enrollee           | -0.067 (0.050)        |
|                                | Cardiac                | -0.096 (0.070)        |
|                                | Cancer                 | 0.445** (0.079)       |
|                                | Neurological           | 0.130 (0.200)         |
|                                | Digestive              | -0.028 (0.076)        |
|                                | Labor                  | -0.002 (0.063)        |
|                                | Newborn baby           | 0.071 (0.087)         |
|                                | Income (\$000)         | 0.005** (0.001)       |
|                                | PPO enrollee           | -0.099* (0.056)       |
| Interactions: For-profit       | Cardiac                | -0.164 (0.181)        |
|                                | Cancer                 | -0.197 (0.202)        |
|                                | Neurological           | 0.229 (0.379)         |
|                                | Digestive              | 0.195 (0.150)         |
|                                | Labor                  | 0.300** (0.107)       |
|                                | Newborn baby           | 0.194* (0.122)        |
|                                | Income (\$000)         | -0.001 (0.003)        |
|                                | PPO enrollee           | -0.036 (0.090)        |
| Interactions: Cardiac services | Cardiac                | 1.222** (0.134)       |
|                                | Income (\$000)         | 0.001 (0.001)         |
|                                | PPO enrollee           | 0.080 (0.088)         |
| Interactions: Imaging services | Cardiac                | -0.188** (0.094)      |
|                                | Cancer                 | -0.052 (0.107)        |
|                                | Neurological           | -0.084 (0.287)        |
|                                | Digestive              | -0.182* (0.105)       |
|                                | Labor                  | -0.071 (0.084)        |
|                                | Newborn baby           | 0.398** (0.129)       |
|                                | Income (\$000)         | 0.004** (0.001)       |
|                                | PPO enrollee           | -0.061 (0.072)        |
| Interactions: Cancer services  | Cancer                 | 0.073 (0.082)         |
|                                | Income (\$000)         | -0.005** (0.001)      |
|                                | PPO enrollee           | 0.087 (0.056)         |
| Interactions: Labor services   | Labor                  | 3.544** (0.391)       |
|                                | Newborn baby           | 3.116** (0.487)       |
|                                | Income (\$000)         | -0.003* (0.002)       |
|                                | PPO enrollee           | 0.045 (0.077)         |
|                                | Hospital fixed effects | Yes                   |
|                                | Pseudo-R <sup>2</sup>  | 0.43                  |

Notes: Maximum likelihood estimation of demand for hospitals using a multinomial logit model. Specification includes hospital fixed effects.  $N = 28666$  encounters. Standard errors are reported in parentheses. \*\* Significant at  $p = 0.05$ ; \* Significant at  $p = 0.1$ .

- This just MLE on the full individual data from Ho (2006)
  - There is no unobserved heterogeneity, just a deterministic  $\beta(y_i)$  where  $y_i$  are demographics.
- The “FTC model” (Raval, Rosenbaum, Wilson RJE 2022)/ (Raval, Rosenbaum, Tenn EI 2017) groups individuals by income, diagnosis, and zip code and estimates a separate set of  $\beta_{g(i)}$  for each group.
- Hospitals are a bit special: distance  $x_{ij}$  does much of the work (special regressor)
- Price endogeneity not really a concern (?)

$$\min_{w_i} \sum_{j,t} \left( s_{jt} - \sum_i w_i \cdot \sigma_{ijt}^*(\beta_i^*, \mathbf{x}_t) \right)^2 \quad \text{subject to} \quad \sigma_{ijt}^*(\beta_i^*, \mathbf{x}_t) = \frac{e^{\beta_i^* x_{jt}}}{1 + \sum_{j'} e^{\beta_i^* x_{j't}}}$$
$$0 \leq w_i \leq 1, \quad \sum_i w_i = 1$$

1. Draw a large number (thousands?) of  $(\beta_i^*)$  from a prior distribution  $g(\beta_i)$  more dispersed than the true  $f(\beta_i)$
2. Compute individual choice probabilities  $\sigma_{ijt}^*(\beta_i^*)$
3. Estimate above by constrained least squares (non-negative lasso)
4. This produces sparse models. (most  $w_i = 0$ )

Fixed coefficients require EM. See Heiss, Hetzenecker, Osterhaus (JE 2022) for details (and elastic net variant  $\sum_i w_i^2 \leq t$ ).

These estimators are helpful when computing choice probabilities is time-consuming (ie: when choices are dynamic). Some recent examples:

- ▶ Nevo, Turner, Williams (ECMA 2016): Broadband competition
- ▶ Blundell, Gowrisankaran, Langer (AER 2020): EPA regulation

## **Review: “Classic” BLP (1995)**

### **Models**

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Add unobservable error for each  $s_{jt}$  labeled  $\xi_{jt}$ .

$$u_{ijt} = \underbrace{x_{jt}\beta - \alpha p_{jt} + \xi_{jt}}_{\delta_{jt}} + \varepsilon_{ijt}, \quad \sigma_j(\delta_t) = \frac{e^{\delta_{jt}}}{1 + \sum_k e^{\delta_{kt}}}$$

- ▶ The idea is that  $\xi_{jt}$  is observed to the firm when prices are set, but not to us the econometricians.
- ▶ Potentially correlated with price  $\text{Corr}(\xi_{jt}, p_{jt}) \neq 0$
- ▶ But not characteristics  $\mathbb{E}[\xi_{jt} \mid x_{jt}] = 0$ .
  - This allows for products  $j$  to be better than some other product in a way that is not fully explained by differences in  $x_j$  and  $x_k$ .
  - Something about a BMW makes it better than a Peugeot but is not fully captured by characteristics that leads to higher sales and/or higher prices.
  - Consumers agree on its value (vertical component).

Taking logs:

$$\ln \sigma_{0t} = -\log \left( 1 + \sum_k \exp[x_{kt}\beta + \xi_{kt}] \right)$$

$$\ln \sigma_{jt} = [x_{jt}\beta - \alpha p_{jt} + \xi_{jt}] - \log \left( 1 + \sum_k \exp[x_{kt}\beta + \xi_{kt}] \right)$$

$$\underbrace{\ln s_{jt} - \ln s_{0t}}_{\text{Data!}} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$$

Exploit the fact that:

1. Observed shares equal the model's choice probabilities:  $s_{jt} = \sigma_{jt}$  (with no sampling error)
2. We have one  $\xi_{jt}$  for every share  $s_{jt}$  (one to one mapping)

## Inversion: Nested Logit (Berry 1994 / Cardell 1991)

This takes a bit more algebra but not much

$$\underbrace{\ln s_{jt} - \ln s_{0t} - \rho \log(s_{j|gt})}_{\text{data!}} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$$

$$\ln s_{jt} - \ln s_{0t} = x_{jt}\beta - \alpha p_{jt} + \rho \log(s_{j|gt}) + \xi_{jt}$$

- ▶ Same as logit plus an extra term  $\log(s_{j|g})$  the **within group share**.
  - We now have a second endogenous regressor.
  - If you don't see it – realize we are regressing  $Y$  on a function of  $Y$ . This should always make you nervous.
- ▶ If you forget to instrument for  $\rho$  you will get  $\rho \rightarrow 1$  because of **attenuation bias**.
- ▶ A common instrument for  $\rho$  is the number of products within the nest. Why?

## What about Zero Shares?

Every inversion so far needs  $\ln s_{jt}$  — but real data are full of  $s_{jt} = 0$ .

- ▶ Weekly store-level scanner data: most UPCs sell zero units in most store-weeks.
- ▶ **Sampling zeros**: finite number of consumers  $N_t$ ; true  $\sigma_{jt} > 0$  but nobody happened to buy.
- ▶ **Structural zeros**: product genuinely not available (or not considered) in market  $t$ .
- ▶ Common hacks and why they fail:
  - Drop the zeros → **selection on  $\xi_{jt}$** : you keep only observations with favorable demand shocks; biases price coefficients toward zero.
  - Replace  $s_{jt}$  with  $s_{jt} + \epsilon$  or Laplace shares  $\frac{n_{jt}+1}{N_t+J_t+1}$  → arbitrary, and elasticities can be badly biased when zeros are common.

## Zeros: What can we do?

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- ▶ Gandhi, Lu, Shi (QE 2023): treat the true  $\sigma_{jt}$  as bounded, construct moment inequalities/bounds that are robust to sampling zeros.
- ▶ Dubé, Hortaçsu, Joo (2021): pairwise-differencing / GMM approach that avoids taking logs of zero shares.
- ▶ Quan and Williams (RAND 2018): aggregate many small local markets (where the zeros live) to a level where shares are positive, modeling the heterogeneity you aggregate over.
- ▶ Practical advice:
  - Zeros are **informative, not missing** — think about *why* the share is zero (availability? consideration? sampling?).
  - Aggregation (across weeks/stores) is often the honest fix.
  - If zeros are structural, model the choice set (product availability, à la Conlon and Mortimer (2013)).

Think about a **generalized inverse** for  $\sigma_j(\delta_t, \mathbf{x}_t, \theta_2) = s_{jt}$  so that

$$\sigma_{jt}^{-1}(\mathbf{s}_t, \tilde{\theta}_2) = \delta_{jt} \equiv x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$$

- ▶ After some transformation of data (shares  $\mathbf{s}_t$ ) we get **mean utilities**  $\delta_{jt}$ .
- ▶ Same IV-GMM approach after transformation
- ▶ Examples:
  - Plain logit and nested logit: the inversions we just derived.
  - Three level nested logit:  $\sigma_j^{-1}(\mathbf{s}_t, \rho) = \ln s_{jt} - \ln s_{0t} - \sum_{d=1}^2 \rho_d \ln \left( \frac{s_{jt}}{s_{d(j),t}} \right)$  (Verboven 1996)
- ▶ Anything with a share requires an IV.

## Aside: Other Analytic Inverses?

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Fosgerau, Monardo, De Palma (2022) propose the IPDL

$$\ln \left( \frac{s_{jt}}{s_{0t}} \right) = \mathbf{x}_{jt} \boldsymbol{\beta} - \alpha p_{jt} + \sum_{d=1}^D \rho_d \ln \left( \frac{s_{jt}}{s_{d(j),t}} \right) + \xi_{jt}$$

- ▶ Can accomodate multiple (partially) overlapping nests
- ▶ The naive idea (include share within group on RHS with IV) actually works like you would want (!)
- ▶ We need that  $\rho > 0$  for this to be RUM.
- ▶ This can allow for mild complementarities as well.

## Aside: The Pure Characteristics Model (Berry and Pakes 2007)

Mixed logit **without** the idiosyncratic taste shock:

$$u_{ijt} = \delta_{jt} + \mu_{ijt} + \underbrace{a}_{\rightarrow 0} \cdot \varepsilon_{ijt}$$

- ▶ Utility comes from characteristics only: vertical tastes over several dimensions (everyone prefers more MHz, more RAM; differ in WTP).
- ▶ Substitution is purely **local** in characteristic space: you compete only with nearby products; allows for **crowding** of the product space.
- ▶ Inversion/estimation much harder: no closed form for  $\sigma_j$ , and the share function is not smooth/invertible the way the logit's is (homotopy: Song; MPCC: Su)  $\rightarrow$  why people avoid it in practice.
- ▶ The logit  $\varepsilon$  adds a “cushion”: it guarantees full support (some substitution to every product) — but also mechanical CS gains from every new product.



## Aside: The Vertical Model (Bresnahan 1987)

All consumers agree on the quality ranking; they differ only in willingness-to-pay for quality:

$$u_{ij} = \delta_j - \alpha_i p_j, \quad \text{sort } p_1 < \dots < p_J \implies \delta_1 < \dots < \delta_J \text{ (why?)}$$

- Analytic inversion via **indifference cutoffs**: choose  $j$  IFF

$$\frac{\delta_{j+1} - \delta_j}{p_{j+1} - p_j} < \alpha_i < \frac{\delta_j - \delta_{j-1}}{p_j - p_{j-1}}, \quad \sigma_j = F\left(\frac{\delta_j - \delta_{j-1}}{p_j - p_{j-1}}\right) - F\left(\frac{\delta_{j+1} - \delta_j}{p_{j+1} - p_j}\right)$$

where  $F(\cdot)$  is the CDF of  $\alpha_i$ .

- Each product competes **only with its quality neighbors**: changing  $p_j$  moves only  $(\sigma_{j-1}, \sigma_j, \sigma_{j+1})$ ; all other cross-elasticities are zero.
- Original application: autos (Bresnahan's 1955 price war). If a luxury car and a truck have similar prices, expect strange substitution patterns.
- Special case of pure characteristics with one dimension. Rarely used today.

We can't solve for  $\delta_{jt}$  directly this time. We often exploit a trick when  $\beta_i, \nu_i$  is normally distributed:

$$\sigma_j(\boldsymbol{\delta}_t; \mathbf{x}_t; \theta_2) = \int \frac{\exp[\delta_{jt} + \mu_{ij}]}{1 + \sum_k \exp[\delta_{kt} + \mu_{ik}]} f(\boldsymbol{\mu}_i | \theta_2)$$

- ▶ We typically parametrize  $\mu_{ijt} = x_{jt} \cdot [\Pi y_i + \Sigma \nu_i]$  where  $y_i$  are demographics and  $\nu_i$  are unobserved heterogeneity (typically multivariate normal).
- ▶ Label  $\theta_2 = [\Pi, \Sigma, \alpha]$
- ▶ This is a  $J \times J$  system of equations for each  $t$ .
- ▶ It is diagonally dominant.
- ▶ There is a unique vector  $\xi_t$  that solves it for each market  $t$ .

## Lots of ways to solve equations (Conlon Gortmaker 2020)

- ▶ If you can work out  $\frac{\partial \sigma_{jt}}{\partial \delta_{kt}}$  (easy) you can solve this using Newton's Method.
- ▶ BLP prove (not easy) that this is a **contraction mapping**.

$$\delta^{(k)}(\theta) = \delta^{(k-1)}(\theta) + \log(s_{jt}) - \log(\sigma_j(\delta_t^{(k-1)}, \theta))$$

- Practical tip:  $\epsilon_{tol}$  needs to be as small as possible. ( $\approx 10^{-13}$ ).
  - Practical tip: Contraction isn't as easy as it looks:  $\log(\sigma_j(\delta_t^{(k-1)}, \theta))$  requires computing the numerical integral each time (either via quadrature or monte carlo).
- ▶ We can use **accelerated fixed point** techniques (SQUAREM) (see Reynaerts, Varadhan, and Nash 2012).  
[PyBLP default].

We can also solve the following convex program:

$$\min_{\boldsymbol{\delta}} \sum_{i=1}^I w_i \cdot \log \left( \sum_j \exp(\delta_j + \mu_{ij}(\theta_2)) \right) - \sum_k \delta_k \cdot s_k$$

The first-order conditions are given by

$$\sigma_j(\boldsymbol{\delta}, \theta_2) = s_j$$

And the Hessian requires knowledge of  $\frac{\partial \sigma_j}{\partial \delta_k}$

- ▶ We are back to solving the non-linear system of equations, but...
- ▶ Convex programming might be faster/more robust than root finding (why?)
- ▶ This makes the proof of a unique solution trivial...

The model is still defined by CMR  $\mathbb{E}[\xi_{jt} \mid z_{jt}^D] = 0$ . From the outside, in:

- ▶ Outer loop: search over nonlinear parameters  $\theta_2$  to minimize GMM objective:

$$\widehat{\theta}_{BLP} = \arg \min_{\theta_2} (Z' \widehat{\xi}(\theta_2)) W (Z' \widehat{\xi}(\theta_2))'$$

- ▶ Inner Loop:

- Solve for  $\delta$  so that  $\sigma_{jt}(\delta, \theta_2) = s_{jt}$ .
  - Computing  $\sigma_{jt}(\delta, \theta_2)$  requires numerical integration (quadrature or monte carlo).
- We can do IV-GMM to recover  $\hat{\alpha}(\theta_2), \hat{\beta}(\theta_2), \hat{\xi}(\theta_2)$  from the change of variables:

$$\delta_{jt} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$$

- Use  $\widehat{\xi}(\theta_2)$  to construct moment conditions.
- ▶ When we have found  $\widehat{\theta}_{2BLP}$  we can use this to update  $W \rightarrow W(\widehat{\theta}_{2BLP})$  and do 2-stage GMM.

- ▶ with enough observations on the same product it is possible to include fixed effects

$$\delta_{jt}(\theta_2) = x_{jt}\beta - \alpha p_{jt} + \underbrace{\xi_{jt}}_{\xi_j + \xi_t + \Delta\xi_{jt}}$$

- ▶ What does  $\xi_j$  mean in this context?
- ▶ What would  $\xi_t$  mean in this context?
- ▶  $\Delta\xi_{jt}$  is now the structural error term, this changes our identification strategy a little.
  - Good: endogeneity problem less severe.
  - Bad: less variation in IV.

- ▶ BLP give us both a statistical **estimator** and an **algorithm** to obtain estimates.
- ▶ Plenty of other algorithms exist
  - We could solve for  $\delta$  using the contraction mapping, using `fsolve` / Newton's Method / Guess and Check (not a good idea!).
  - We could try and consider a non-nested estimator for the BLP problem instead of solving for  $\delta(\theta_2), \xi(\theta_2)$  we could let  $\delta, \xi, \alpha, \beta$  be free parameters.
- ▶ We could think about different statistical estimators such as  $K$ -step GMM, Continuously Updating GMM, etc.

$$\arg \min_{\theta_2} \psi' \Omega^{-1} \psi \quad \text{s.t.}$$

$$\psi = \xi(\theta_2)' Z$$

$$\xi_{jt}(\theta_2) = \delta_{jt}(\theta_2) - x_{jt}\beta - \alpha p_{jt} \quad (1)$$

$$\log(s_{jt}) = \log(\sigma_{jt}(\delta, \theta_2))$$

$$\arg \min_{\theta_2, \alpha, \beta, \xi, \psi} \psi' \Omega^{-1} \psi \quad \text{s.t.}$$

$$\psi = \xi' Z \quad (2)$$

$$\xi_{jt} = \delta_{jt} - x_{jt}\beta - \alpha p_{jt}$$

$$\log(s_{jt}) = \log(\sigma_{jt}(\theta_2, \delta))$$



- ▶ The original BLP paper and the DFS paper define different **algorithms** to produce the same statistical **estimator**.
  - The BLP algorithm is a **nested fixed point** (NFP) algorithm.
  - The DFS algorithm is a **mathematical program with equilibrium constraints** (MPEC).
  - The unknown parameters satisfy the same set of first-order conditions. (Not only asymptotically, but in finite sample).
  - $\hat{\theta}_{NFP} \approx \hat{\theta}_{MPEC}$  but for numerical differences in the optimization routine.
- ▶ Our choice of algorithm should mostly be about computational convenience.

### ► Advantages

- Concentrate out all of the linear in utility parameters  $(\xi, \delta, \beta)$  so that we only search over  $\theta_2$ . When  $\dim(\Sigma) = \theta_2$  is small (few dimensions of unobserved heterogeneity) this is a big advantage. For  $K \leq 5$  this is my preferred approach.
- When  $T$  (number of markets/periods) is large then you can exploit solving in parallel for  $\delta$  market by market.

### ► Disadvantages

- Small numerical errors in contraction can be amplified in the outer loop,  $\rightarrow$  tolerance needs to be very tight.
- Errors in numerical integration can also be amplified in the outer loop  $\rightarrow$  must use a large number of draws/nodes.
- Hardest part is working out the Jacobian via IFT.

### ► Advantages

- Problem scales better in  $\dim(\theta_2)$ .
- Because all constraints hold at the optimum only: less impact of numerical error in tolerance or integration.
- Derivatives are less complicated than  $\frac{\partial \delta}{\partial \theta_2}$  (no IFT).

### ► Disadvantages

- We are no longer concentrating out parameters, so there are a lot more of them! Storing the (Hessian) matrix of second derivatives can be difficult on memory → Fixed effects are harder
- We have to find the derivatives of the shares with respect to all of the parameters  $\beta, \xi, \theta_2$ . (The other derivatives are pretty easy).
- Parallelizing the derivatives is trickier than NFP case.

## **Adding Supply**

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- ▶ Economic theory gives us some additional powerful restrictions.
- ▶ We may want to impose  $MR = MC$ .
- ▶ Alternatively, we can ask – what is a good instrument for demand? something from another equation (ie: supply).

We can break up the parameter space into three parts:

- ▶  $\theta_1$ : linear exogenous demand parameters,
- ▶  $\theta_2$ : parameters including price and random coefficients (endogenous / nonlinear)
- ▶  $\theta_3$ : linear exogenous supply parameters.

Consider the multi-product Bertrand FOCs:

$$\begin{aligned}\arg \max_{\mathbf{p} \in \mathcal{J}_f} \pi_f(\mathbf{p}) &= \sum_{j \in \mathcal{J}_f} (p_j - c_j) \cdot \sigma_j(\mathbf{p}) + \kappa_{fg} \sum_{k \in \mathcal{J}_g} (p_k - c_k) \cdot \sigma_k(\mathbf{p}) \\ 0 &= \sigma_j(\mathbf{p}) + \sum_{k \in \mathcal{J}_f} (p_k - c_k) \frac{\partial \sigma_k}{\partial p_j}(\mathbf{p})\end{aligned}$$

It is helpful to define the **ownership matrix**:

$$\mathcal{H}(\kappa)_{(j,k)} = \begin{cases} 1 & \text{for } (j,k) \in \mathcal{J}_f \text{ for any } f \\ 0 & \text{o.w} \end{cases}$$

We can re-write the FOC in matrix form where  $\odot$  denotes Hadamard product (element-wise) and

$\Delta_{(j,k)}(\mathbf{p}) = -\frac{\partial \sigma_j}{\partial p_k}(\mathbf{p})$  is the demand derivatives matrix:

$$\begin{aligned}\sigma(\mathbf{p}) &= (\mathcal{H} \odot \Delta(\mathbf{p})) \cdot (\mathbf{p} - \mathbf{mc}), \\ \mathbf{mc} &= \mathbf{p} - \underbrace{(\mathcal{H} \odot \Delta(\mathbf{p}))^{-1} \sigma(\mathbf{p})}_{\eta(\mathbf{p}, \mathbf{s}, \theta_2)}.\end{aligned}$$

## Recovering Marginal Costs

Recover implied markups/ marginal costs, and assume a functional form for  $mc_{jt}(x_{jt}, w_{jt})$ .

$$\begin{aligned}\mathbf{mc}(\theta) &= \mathbf{p} - \boldsymbol{\eta}(\mathbf{p}, \mathbf{s}, \theta_2) \\ f(mc_{jt}) &= [\mathbf{x}_{jt}, \mathbf{w}_{jt}] \theta_3 + \omega_{jt}\end{aligned}$$

Which we can solve for  $\omega_{jt}$ :

$$\omega_{jt} = f(\mathbf{p} - \boldsymbol{\eta}(\mathbf{p}, \mathbf{s}, \theta_2)) - [\mathbf{x}_{jt}, \mathbf{w}_{jt}] \theta_3$$

- ▶  $f(\cdot)$  is usually  $\log(\cdot)$  or identity.
- ▶ We can use this to form additional moments:  $\mathbb{E}[\omega_{jt} \mid z_{jt}^s] = 0$ .
- ▶ We can just stack these up with the demand moments  $E[\xi'_{jt} Z_{jt}^d] = 0$ .
- ▶ This step is optional but can aid in identification (if you believe it).



Some different definitions:

$$\begin{aligned}y_{jt}^D &:= \hat{\delta}_{jt}(\theta_2) + \alpha p_{jt} = (\mathbf{x}_{jt} \mathbf{v}_{jt})' \beta + \xi_t =: \mathbf{x}_{jt}^{D'} \beta + \xi_{jt} \\ y_{jt}^S &:= \widehat{mc}_{jt}(\theta_2) = (\mathbf{x}_{jt} \mathbf{w}_{jt})' \gamma + \omega_t =: \mathbf{x}_{jt}^{S'} \gamma + \omega_{jt}\end{aligned}\tag{3}$$

Stacking the system across observations yields:

$$\underbrace{\begin{bmatrix} y_D \\ y_S \end{bmatrix}}_{2N \times 1} = \underbrace{\begin{bmatrix} X_D & 0 \\ 0 & X_S \end{bmatrix}}_{2N \times (K_1 + K_3)} \underbrace{\begin{bmatrix} \beta \\ \gamma \end{bmatrix}}_{(K_1 + K_3) \times 1} + \underbrace{\begin{bmatrix} \xi \\ \omega \end{bmatrix}}_{2N \times 1}\tag{4}$$

## Simultaneous Supply and Demand: in details

- (a) For each market  $t$ : solve  $s_{jt} = \sigma_{jt}(\delta_{.t}, \theta_2)$  for  $\hat{\delta}_{.t}(\theta_2)$ .
- (b) For each market  $t$ : use  $\hat{\delta}_{.t}(\theta_2)$  to construct  $\eta_{.t}(\mathbf{q}_t, \mathbf{p}_t, \hat{\delta}_{.t}(\theta_2), \theta_2)$
- (c) For each market  $t$ : Recover  $\widehat{mc}_{jt}(\hat{\delta}_{.t}(\theta_2), \theta_2) = p_{jt} - \eta_{jt}(\hat{\delta}_{.t}(\theta_2), \theta_2)$
- (d) Stack up  $\hat{\delta}_{.t}(\theta_2)$  and  $\widehat{mc}_{jt}(\hat{\delta}_{.t}(\theta_2), \theta_2)$  into the stacked system on the previous slide and use linear IV-GMM to recover  $[\hat{\theta}_1(\theta_2), \hat{\theta}_3(\theta_2)]$  following the recipe in Appendix of Conlon Gortmaker (2020)
- (e) Construct the residuals  $\hat{\xi}_{jt}(\theta_2)$  and  $\hat{\omega}_{jt}(\theta_2)$  from the stacked system.
- (f) Construct sample moments

$$g_n^D(\theta_2) = \frac{1}{N} \sum_{jt} Z_{jt}^{D'} \hat{\xi}_{jt}(\theta_2)$$

$$g_n^S(\theta_2) = \frac{1}{N} \sum_{jt} Z_{jt}^{S'} \hat{\omega}_{jt}(\theta_2)$$

(g) Construct GMM objective  $Q_n(\theta_2) = \begin{bmatrix} g_n^d(\theta_2) \\ g_n^s(\theta_2) \end{bmatrix}' W \begin{bmatrix} g_n^d(\theta_2) \\ g_n^s(\theta_2) \end{bmatrix}$

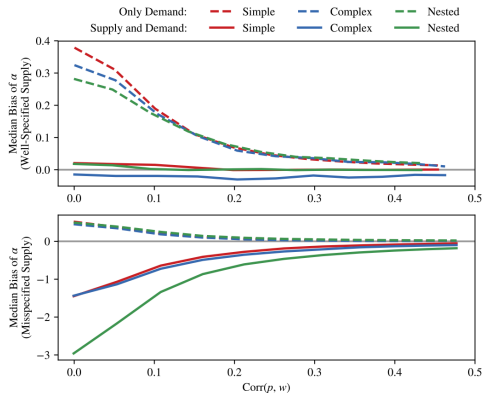
## What's the Point (Conlon Gortmaker RJE 2020)

- ▶ A well-specified supply side can make it easier to estimate  $\theta_2$  parameters (price in particular).
- ▶ Imposing the supply side only helps if we have information about the marginal costs / production function that we would like to impose
- ▶ May want to enforce some economic constraints: ( $mc_{jt} > 0$  is a good one).
- ▶ But assuming the wrong conduct  $\mathcal{H}_t$  can lead to misspecification!

| Simulation | Supply | Instruments | Seconds | True Value |            |            |        | Median Bias |            |            |        | Median Absolute Error |            |            |        |
|------------|--------|-------------|---------|------------|------------|------------|--------|-------------|------------|------------|--------|-----------------------|------------|------------|--------|
|            |        |             |         | $\alpha$   | $\sigma_x$ | $\sigma_p$ | $\rho$ | $\alpha$    | $\sigma_x$ | $\sigma_p$ | $\rho$ | $\alpha$              | $\sigma_x$ | $\sigma_p$ | $\rho$ |
| Simple     | No     | Own         | 0.6     | -1         | 3          |            |        | 0.126       | -0.045     |            |        | 0.238                 | 0.257      |            |        |
| Simple     | No     | Sums        | 0.6     | -1         | 3          |            |        | 0.224       | -0.076     |            |        | 0.257                 | 0.208      |            |        |
| Simple     | No     | Local       | 0.6     | -1         | 3          |            |        | 0.181       | -0.056     |            |        | 0.242                 | 0.235      |            |        |
| Simple     | No     | Quadratic   | 0.6     | -1         | 3          |            |        | 0.206       | -0.085     |            |        | 0.263                 | 0.239      |            |        |
| Simple     | No     | Optimal     | 0.8     | -1         | 3          |            |        | 0.218       | -0.049     |            |        | 0.250                 | 0.174      |            |        |
| Simple     | Yes    | Own         | 1.4     | -1         | 3          |            |        | 0.021       | 0.006      |            |        | 0.226                 | 0.250      |            |        |
| Simple     | Yes    | Sums        | 1.5     | -1         | 3          |            |        | 0.054       | -0.020     |            |        | 0.193                 | 0.196      |            |        |
| Simple     | Yes    | Local       | 1.4     | -1         | 3          |            |        | 0.035       | -0.006     |            |        | 0.207                 | 0.229      |            |        |
| Simple     | Yes    | Quadratic   | 1.4     | -1         | 3          |            |        | 0.047       | -0.022     |            |        | 0.217                 | 0.237      |            |        |
| Simple     | Yes    | Optimal     | 2.2     | -1         | 3          |            |        | 0.005       | 0.012      |            |        | 0.170                 | 0.171      |            |        |

# What about Misspecification? (Conlon Gortmaker RJE 2020)

Figure 2: Instrument Strength and Misspecification



Each plot documents how bias of the linear parameter on price,  $\alpha$ , decreases with the strength of the cost shifter  $w_{jt}$ , which is included as a demand-side instrument. To weaken or strengthen the instrument, we vary its supply-side parameter from  $\gamma_w = 0$  to  $\gamma_w = 1$ , and report the correlation this induces between  $w_{jt}$  and prices  $p_{jt}$ . Reported bias values are medians across 1,000 different simulations. The top plot reports results for the simulation configurations described in Section 5. In the bottom plot, we simulate data according to perfect competition (i.e., prices are set equal to marginal costs instead of those that satisfy Bertrand-Nash first order conditions), but continue to estimate the model under the assumption of imperfect competition. For all problems, we use the “approximate” version of the feasible optimal instruments and a Gauss-Hermite product rule that exactly integrates polynomials of degree 17 or less.

## Quick Case Study

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## What's the point?

$$\mathbf{mc} = \mathbf{p} - \eta(\mathbf{p}, \mathbf{s}, \theta_2).$$

Demand systems have two main deliverables:

- ▶ Own-price elasticities  $\epsilon_{jj}(\mathbf{p}, \theta_2)$
- ▶ Substitution patterns (all transforms of same object)
  - Demand Derivatives:  $\Delta(\mathbf{p}, \theta_2)$
  - Cross elasticities:  $\epsilon_{jk}(\mathbf{p}) = \frac{\partial q_k}{\partial p_j}$
  - Diversion Ratios:  $D_{jk}(\mathbf{p}) = \frac{\partial q_k}{\partial p_j} / \left| \frac{\partial q_j}{\partial p_j} \right|$
- ▶ Other checks:  $D_{j0}(\mathbf{p})$  diversion to outside good;  $\epsilon^{agg}$  category elasticity to 1% tax.
  - Tend to give insight into the economics of the problem.

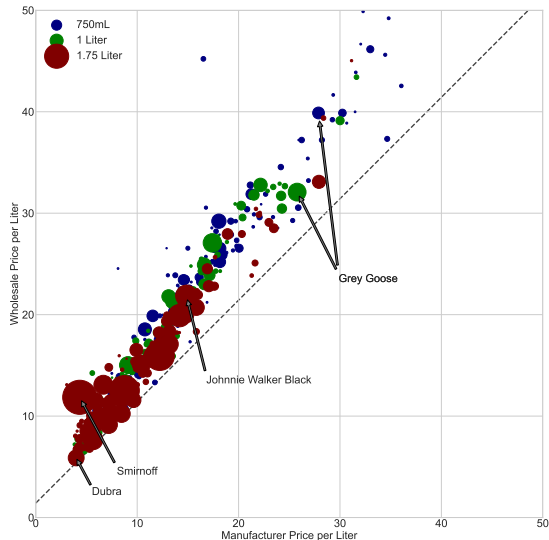
## Why does supply matter? (Conlon Rao 2025)

Consumer  $i$  chooses product  $j$  (brand-size-flavor) in quarter  $t$ :

$$u_{ijt} = \beta_i^0 - \alpha_i p_{jt} + \beta_i^{1750} \cdot \mathbb{I}[1750mL]_j + \gamma_j + \gamma_t + \varepsilon_{ijt}(\rho)$$
$$\begin{pmatrix} \ln \alpha_i \\ \beta_i \end{pmatrix} = \begin{pmatrix} \bar{\alpha} \\ \theta_1 \end{pmatrix} + \Sigma \cdot \nu_i + \sum_k \Pi_k \cdot \mathbb{I}\{LB_k \leq \text{Income}_i < UB_k\}$$

- ▶ Nesting Parameter  $\rho$ : Substitution within category (Vodka, Gin, etc.)
- ▶ Consumers of different income levels have different mean values for coefficients
- ▶ Conditional on income, normally distributed unobserved heterogeneity for:
  - Price  $\alpha_i$
  - Constant  $\beta_i^0$  (Overall demand for spirits)
  - Package Size:  $\beta_i^{1750}$  (Large vs. small bottles)

# Wholesale Margins Under Post and Hold



- ▶ Price Cost Margins (and Lerner Markups) are higher on premium products
- ▶ Markups on least expensive products (plastic bottle vodka) are very low.
- ▶ Smirnoff (1.75L) is best seller (high markup / outlier).
- ▶ A planner seeking to minimize ethanol consumption would flatten these markups!
- ▶ Matching this pattern is kind of the whole ballgame !
- ▶ Plain logit gives  $\epsilon_{jj} = \alpha \cdot p_j \cdot (1 - \sigma_j)$ .

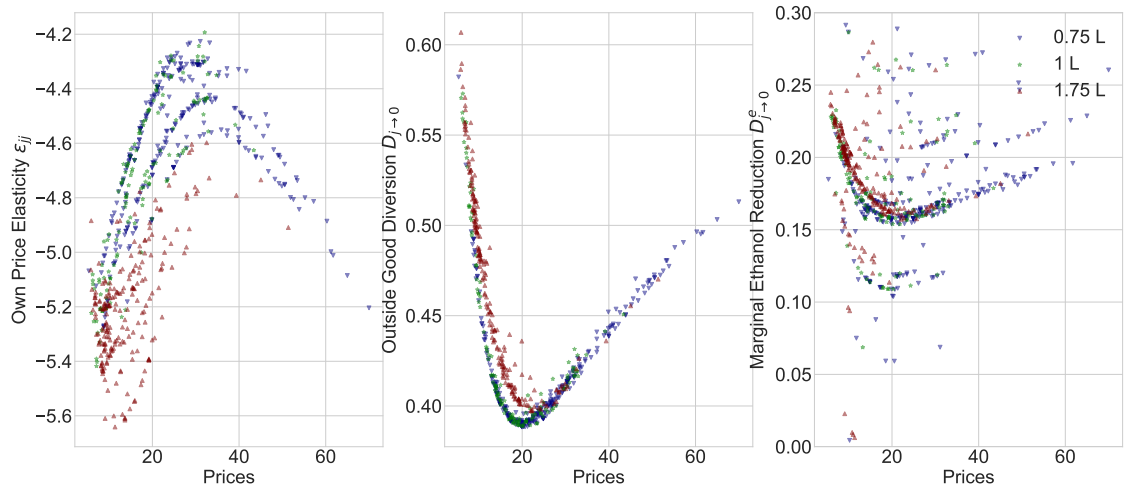


# Demand Estimates (from PyBLP, Conlon Gortmaker (2020, 2023))

| II                              | Const             | Price             | 1750mL            |
|---------------------------------|-------------------|-------------------|-------------------|
| Below \$25k                     | 2.433<br>(0.287)  | -0.736<br>(0.056) | -0.442<br>(0.083) |
| \$25k-\$45k                     | 0.243<br>(0.328)  | -0.720<br>(0.095) | -0.258<br>(0.097) |
| \$45k-\$70k                     | 0.000<br>(0.000)  | -0.768<br>(0.094) | 0.000<br>(0.000)  |
| \$70k-\$100k                    | -0.960<br>(0.324) | -1.032<br>(0.094) | -0.275<br>(0.096) |
| Above \$100k                    | -3.762<br>(0.262) | -2.291<br>(0.074) | -0.794<br>(0.077) |
| $\Sigma^2$                      |                   |                   |                   |
| Const                           | 3.868<br>(0.740)  | 1.271<br>(0.150)  | 0.000<br>(0.000)  |
| Price                           | 1.271<br>(0.150)  | 0.418<br>(0.031)  | 0.000<br>(0.000)  |
| Nesting Parameter $\rho$        | 0.27<br>(0.021)   |                   |                   |
| Fixed Effects                   | Brand+Quarter     |                   |                   |
| Model Predictions               | 25%               | 50%               | 75%               |
| Own Elasticity                  | -5.072            | -4.772            | -4.484            |
| Aggregate Elasticity            | -0.545            | -0.530            | -0.506            |
| Observed Wholesale Markup (PH)  | 0.188             | 0.233             | 0.276             |
| Predicted Wholesale Markup (PH) | 0.222             | 0.238             | 0.255             |
| Outside Good Diversion $D_{j0}$ | 0.428             | 0.458             | 0.496             |
| Ethanol Diversion $D_{j0}^e$    | 0.171             | 0.188             | 0.211             |

- Demographic Interactions w/ 5 income bins (matched to micro-moments: defined properly next lecture)
- Correlated Normal Tastes: (Constant, Large Size, Price)
- Supply moments exploit observed upstream prices and tax change (ie: match observed markups).  
$$\mathbb{E}[\omega_{jt}] = 0, \text{ with } \omega_{jt} = (p_{jt}^w - p_{jt}^m - \tau_{jt}) - \eta_{jt}(\theta_2).$$
- Match re-purchase rate of *Vodka*  $\rightarrow$  *Vodka* for each category.
- Pass-through consistent with estimates from our AEJ:Policy paper.

# Elasticities and Diversion Ratios



# Diversion Ratios

|                                       | Median Price | % Substitution |                                  | Median Price | % Substitution |
|---------------------------------------|--------------|----------------|----------------------------------|--------------|----------------|
| Capt Morgan Spiced 1.75 L (\$15.85)   |              |                | Cuervo Gold 1.75 L (\$18.33)     |              |                |
| Bacardi Superior Lt Dry Rum 1.75 L    | 12.52        | 7.59           | Cuervo Gold 1.0 L                | 21.32        | 3.26           |
| Bacardi Superior Lt Dry Rum 1.0 L     | 15.03        | 2.06           | Sauza Giro Tequila Gold 1.0 L    | 8.83         | 2.15           |
| Smirnoff 1.75 L                       | 11.85        | 1.87           | Don Julio Silver 1.75 L          | 22.81        | 2.12           |
| Bacardi Dark Rum 1.75 L               | 12.52        | 1.57           | Smirnoff 1.75 L                  | 11.85        | 1.80           |
| Lady Bligh Spiced V Island Rum 1.75 L | 9.43         | 1.46           | Cuervo Gold 0.75 L               | 23.44        | 1.44           |
| Woodford 0.75 L (\$34.55)             |              |                | Beefeater Gin 1.75 L (\$17.09)   |              |                |
| Jack Daniel Black Label 1.0 L         | 27.08        | 4.25           | Tanqueray 1.75 L                 | 17.09        | 7.11           |
| Jack Daniel Black Label 1.75 L        | 21.85        | 4.19           | Gordons 1.75 L                   | 11.19        | 2.55           |
| Jack Daniel Black Label 0.75 L        | 29.21        | 2.66           | Seagrams Gin 1.75 L              | 10.23        | 1.84           |
| Makers Mark 1.0 L                     | 32.79        | 2.46           | Smirnoff 1.75 L                  | 11.85        | 1.82           |
| Makers Mark 0.75 L                    | 31.88        | 1.53           | Gilbey Gin 1.75 L                | 9.30         | 1.56           |
| Dubra Vdk Dom 80P 1.75 L (\$5.88)     |              |                | Belvedere Vodka 0.75 L (\$30.55) |              |                |
| Popov Vodka 1.75 L                    | 7.66         | 3.88           | Absolut Vodka 1.75 L             | 15.94        | 3.34           |
| Smirnoff 1.75 L                       | 11.85        | 2.79           | Grey Goose 1.0 L                 | 32.08        | 2.71           |
| Sobieski Poland 1.75 L                | 9.09         | 1.93           | Smirnoff 1.75 L                  | 11.85        | 2.36           |
| Grays Peak Vdk Dom 1.75 L             | 9.16         | 1.78           | Ktl1 Vdk Im 1.75 L               | 20.71        | 1.49           |
| Bellows Vodka 1.0 L                   | 6.21         | 1.49           | Absolut Vodka 1.0 L              | 24.91        | 1.47           |

## **Instruments and Identification**

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- Once we have  $\delta_{jt}(\theta)$  identification of linear parameters  $\theta_1 = [\beta, \xi_j, \xi_t]$  is pretty straightforward

$$\delta_{jt}(\theta) = x_{jt}\beta - \alpha p_{jt} + \xi_j + \xi_t + \Delta\xi_{jt}$$

- This is either basic linear IV or panel linear IV.
- Intuition: How are  $\theta_2$  taste parameters identified?
- Consider increasing the price of  $j$  and measuring substitution to other products  $k, k'$  etc.
  - If sales of  $k$  increase with  $p_j$  and  $(x_j^{(1)}, x_k^{(1)})$  are similar then we increase the  $\theta_2$  that corresponds to  $x^{(1)}$ .
  - Price is the most obvious to vary, but sometimes this works for other characteristics (like distance).
  - Alternative: vary the set of products available to consumers by adding or removing an option.

- ▶ Recall the nested logit, where there are two separate endogeneity problems
  - **Price**: this is the familiar one!
  - **Nonlinear characteristics/Shares**  $\theta_2$  this is the other one.
- ▶ We are doing nonlinear GMM: Start with  $\mathbb{E}[\xi_{jt}|x_{jt}, z_{jt}] = 0$  use  $\mathbb{E}[\xi' [ZX]] = 0$ .
  - In practice this means that for valid instruments  $(x, z)$  any function  $f(x, z)$  is also a valid instrument  $\mathbb{E}[\xi_{jt} f(x_{jt}, z_{jt})] = 0$ .
  - We can use  $x, x^2, x^3, \dots$  or interactions  $x \cdot z, x^2 \cdot z^2, \dots$
  - What is a reasonable choice of  $f(\cdot)$ ?
  - Where does  $z$  come from?

## Exclusion Restrictions (see Berry Haile 2014)

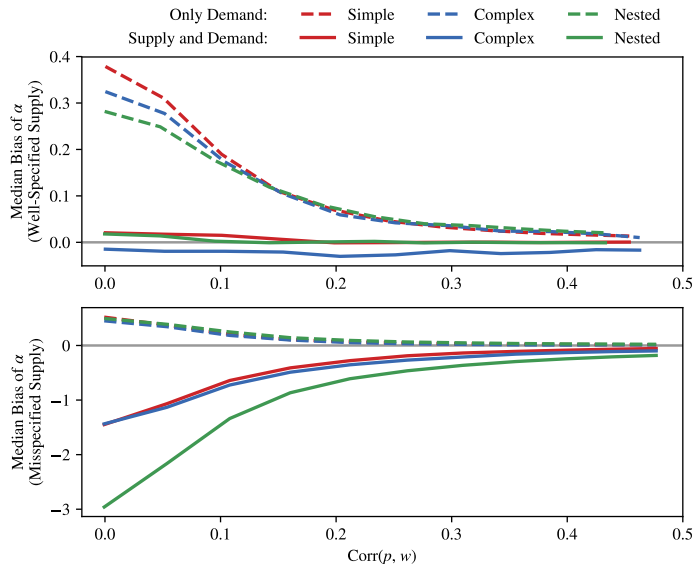
$$\begin{aligned}\delta_{jt}(\mathbf{s}_t, \mathbf{y}_t, \tilde{\theta}_2) &= [\mathbf{x}_{jt}, \mathbf{v}_{jt}] \beta - \alpha p_{jt} + \xi_{jt} \\ f(p_{jt} - \eta_{jt}(\theta_2, \mathbf{p}, \mathbf{s})) &= h(\mathbf{x}_{jt}, \mathbf{w}_{jt}; \theta_3) + \omega_{jt}\end{aligned}$$

The first place to look for exclusion restrictions/instruments:

- ▶ Something in another equation!
- ▶  $\mathbf{v}_j$  shifts demand but not supply
- ▶  $\mathbf{w}_j$  shifts supply but not demand
- ▶  $\mathbf{y}_t$  is a sneaky demand shifter
- ▶ If it doesn't shift either is it really relevant?

Alternative: MacKay Miller (2022) propose  $Cov(\xi_{jt}, \omega_{jt}) = 0$  as an alternative.

## Cost Shifters Really Matter (from Conlon Gortmaker RJE)





## What about Hausman Instruments?

AKA contemporaneous prices of same product in a different market.

- ▶ Idea is to pick up common cost shocks:

$$p_{jmt} = c_{jmt} + \eta_{jmt}$$

- ▶ But this places strong assumptions on nature of demand shocks (and markups  $\eta_{jmt}$ )
- ▶ Even with FE:  $\xi_{jmt} = \xi_j + \xi_t + \underbrace{\Delta\xi_{jt}}_{=0} + \Delta\xi_{jmt}$
- ▶ A common complaint: national advertising might increase demand for a product in multiple geographic markets.

The equilibrium markup is a function of **everything!**  $\eta_{jt}(\mathbf{p}, \mathbf{s}, \xi_t, \omega_t, \mathbf{x}_t, \mathbf{w}_t, \mathbf{v}_t, \mathbf{y}_t, \theta_2)$ :

- ▶ It is obviously **endogenous** (depends on error terms)!
- ▶ But lots of potential instruments beyond **excluded**  $\mathbf{v}_t$  or  $\mathbf{w}_t$ .
- ▶ Idea: cross-market variation in number or strength of competitors
  - Also  $\mathbf{v}_{-j}$  and  $\mathbf{w}_{-j}$  and  $\mathbf{x}_{-j}$ .
  - Not  $p_{-j}$  or  $\xi_{-j}$ , etc.
  - The idea is that these instruments shift the **marginal revenue curve**.
  - What is a good choice of  $f(\mathbf{x}_{-j})$ ? etc.

- ▶ Common choices are average characteristics of other products in the same market  $f(x_{-j,t})$ . **BLP instruments**
  - Same firm  $z_{1jt} = \bar{x}_{-j_f,t} = \frac{1}{|F_j|} \sum_{k \in \mathcal{F}_j} x_{kt} - \frac{1}{|F_j|} x_{jt}$ .
  - Other firms  $z_{2jt} = \bar{x}_{\cdot,t} - \bar{x}_{-j_f,t} - \frac{1}{J} x_{jt}$ .
  - Plus regressors  $(1, x_{jt})$ .
  - Plus higher order interactions
- ▶ Technically linearly independent for large (finite)  $J$ , but becoming highly correlated.
  - Can still exploit variation in number of products per market or number of products per firm.
- ▶ Correlated moments  $\rightarrow$  “many instruments”.
  - May be inclined to “fix” correlation in instrument matrix directly.

Consider the limit as  $J \rightarrow \infty$

$$\frac{\sigma_{jt}(\mathbf{p}_t)}{\left| \frac{\partial \sigma_{jt}(\mathbf{p}_t)}{\partial p_{jt}} \right|} = \frac{1}{\alpha} \frac{1}{1 - \sigma_{jt}} \rightarrow \frac{1}{\alpha}$$

- ▶ Hard to use markup shifting instruments to instrument for a constant.
- ▶ How close to the constant do we get in practice?
- ▶ Average of  $x_{-j}$  seems like an especially poor choice. Why?
- ▶ Shows there may still be some power in: products per market, products per firm.
- ▶ Convergence to constant extends to mixed logits (see Gabaix and Laibson 2004).
- ▶ Suggests that you really need cost shifters.

- ▶ Also need instruments for the  $\Sigma$  random coefficient parameters.
- ▶ Instead of average of other characteristics  $f(x) = \frac{1}{J-1} \sum_{k \neq j} x_k$ , can transform as distance to  $x_j$ .

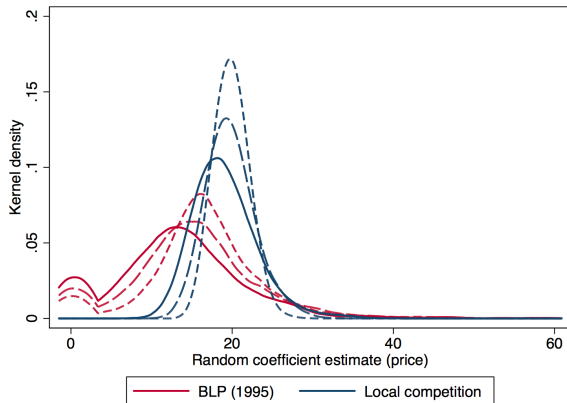
$$d_{jt}^k = |x_k - x_j|$$

- ▶ And use this transformed to construct two kinds of IV (Squared distance, and count of local competitors)

$$DIV_1 = \sum_{j \in F} d_{jt}^2, \quad \sum_{j \notin F} d_{jt}^2$$
$$DIV_2 = \sum_{j \in F} \mathbb{I}[d_{jt} < c] \quad \sum_{j \notin F} \mathbb{I}[d_{jt} < c]$$

- ▶ They choose  $c$  to correspond to one standard deviation of  $x$  across markets.
- ▶ Monotonicity?

Figure 4: Distribution of parameter estimates in small and large samples



**Sample size:** Solid = 500, Long dash = 1,000, Dash = 2,500.

## Intuition from Linear IV (FRAC: Salanie and Wolak)

Simple case where  $\theta_0 = (\beta_0, \Pi_0, \Sigma_0)'$ . A second-order Taylor expansion around  $\Pi_0 = \Sigma_0 = 0$  gives the following linear model with four regressors:

$$\log \frac{s_{jt}}{s_{0t}} \approx \beta_0 x_{jt} + \Sigma_0^2 a_{jt} + \Pi_0 m_t^y x_{jt} + \Pi_0^2 v_t^y a_{jt} + \xi_{jt}, \quad a_{jt} = \left( \frac{x_{jt}}{2} - \sum_{k \in \mathcal{J}_t} s_{kt} \cdot x_{kt} \right) \cdot x_{jt} \quad (5)$$

- ▶  $m_t^y = \sum_{i \in \mathcal{J}_t} w_{it} \cdot y_{it}$  is the within-market demographic mean
- ▶  $v_t^y = \sum_{i \in \mathcal{J}_t} w_{it} \cdot (y_{it} - m_t^y)^2$  is its variance
- ▶  $a_{jt}$  is an “artificial regressor” that reflects within-market differentiation of the product characteristic  $x_{jt}$ .
- ▶ Linear but we still need an IV for  $a_{jt}$ .

Implemented in Julia by Jimbo Brand <https://github.com/jamesbrandecon/FRAC.jl>

## Connection or when do GH IV work well?

Recall the GH IV are:

$$J \cdot x_{jt}^2 + \underbrace{\sum_k x_{kt}^2}_{\text{constant for } t} - 2 \sum_k x_{jt} \cdot x_{kt}$$

and the artificial regressor is

$$\frac{1}{2}x_{jt}^2 - 2x_{jt} \cdot \sum_k s_{kt} \cdot x_{kt}$$

- ▶ We should be **share weighting** the interaction term, but GH assume equal weighting.
- ▶ Should be able to do better than these IV (but ideal is infeasible...)
- ▶ Alternative take: GH propose IIA test that looks a lot like Salanie Wolak estimator. Good for starting values?  
Or as pre-test for heterogeneity?
- ▶ Warning: I find these are always nearly colinear and run PCA first...



Chamberlain (1987) asks how can we choose  $f(z_i)$  to obtain the semi-parametric efficiency bound with conditional moment restrictions:

$$\mathbb{E}[g(z_i, \theta) | z_i] = 0 \Rightarrow \mathbb{E}[g(z_i, \theta) \cdot f(z_i)] = 0$$

The answer is to choose instruments related to the (expected) Jacobian of moment conditions w.r.t  $\theta$ . The true Jacobian at  $\theta_0$  is **infeasible**:

$$G = \mathbb{E} \left[ \frac{\partial g(z_i, \theta)}{\partial \theta} | z_i, \theta_0 \right]$$

Consider the simplest IV problem:

$$y_i = \beta x_i + \gamma v_i + u_i \quad \text{with} \quad \mathbb{E}[u_i | v_i, z_i] = 0$$

$$u_i = (y_i - \beta x_i - \gamma v_i)$$

$$g(x_i, v_i, z_i) = (y_i - \beta x_i - \gamma v_i)$$

Which gives:

$$\mathbb{E} \left[ \frac{\partial g(x_i, v_i, z_i, \theta)}{\partial \gamma} \mid v_i, z_i \right] \propto v_i$$

$$\mathbb{E} \left[ \frac{\partial g(x_i, v_i, z_i, \theta)}{\partial \beta} \mid v_i, z_i \right] \propto \mathbb{E}[x_i \mid v_i, z_i]$$

We can't just use  $x_i$  (bc endogenous!), but you can also see where 2SLS comes from...

From previous slide, nothing says that  $\mathbb{E}[x_i \mid v_i, z_i]$  needs to be **linear**!

- ▶ Since any  $f(x, z)$  satisfies our orthogonality condition, we can try to choose  $f(x, z)$  as a **basis** to approximate optimal instruments.
- ▶ Why? Well affine transformations of instruments are still valid, and we span the same vector space!
- ▶ We are essentially relying on a non-parametric regression that we never run (but could!)
  - This is challenging in practice – and in fact suffers from a curse of dimensionality.
  - This is frequently given as a rationale behind higher order  $x$ 's.
  - When the dimension of  $x$  is low – this may still be feasible. ( $K \leq 5$ ).
  - But recent improvements in sieves, LASSO, non-parametric regression are encouraging.

Recall the GMM moment conditions are given by  $\mathbb{E}[\xi_{jt}|Z_{jt}^D] = 0$  and  $\mathbb{E}[\omega_{jt}|Z_{jt}^S] = 0$  and the asymptotic GMM variance depends on  $(G' \Omega^{-1} G)$  where the expressions are given below:

$$G = \mathbb{E} \left[ \left( \frac{\partial \xi_{jt}}{\partial \theta}, \frac{\partial \omega_{jt}}{\partial \theta} \right) | \mathbf{Z}_t \right], \quad \Omega = \mathbb{E} \left[ \begin{pmatrix} \xi_{jt} \\ \omega_{jt} \end{pmatrix} \begin{pmatrix} \xi_{jt} & \omega_{jt} \end{pmatrix} | \mathbf{Z}_t \right].$$

Chamberlain (1987) showed that the approximation to the optimal instruments are given by the expected Jacobian contribution for each observation  $(j, t)$ :  $\mathbb{E}[G_{jt}(\mathbf{Z}_t) \Omega_{jt}^{-1} | \mathbf{Z}_t]$ .

## Optimal Instruments (see Conlon Gortmaker 2020)

BLP 1999 tells us the (Chamberlain 1987) optimal instruments for this supply-demand system of  $G \Omega^{-1}$  where for a given observation  $n$ , we need to compute  $\mathbb{E}[\frac{\partial \xi_{jt}}{\partial \theta} | \mathbf{Z}_t]$  and  $\mathbb{E}[\frac{\partial \omega_{jt}}{\partial \theta} | \mathbf{Z}_t]$

$$G_{jt} \equiv \underbrace{\begin{bmatrix} \frac{\partial \xi_{jt}}{\partial \beta} & \frac{\partial \omega_{jt}}{\partial \beta} \\ \frac{\partial \xi_{jt}}{\partial \alpha} & \frac{\partial \omega_{jt}}{\partial \alpha} \\ \frac{\partial \xi_{jt}}{\partial \theta_2} & \frac{\partial \omega_{jt}}{\partial \theta_2} \\ \frac{\partial \xi_{jt}}{\partial \gamma} & \frac{\partial \omega_{jt}}{\partial \gamma} \end{bmatrix}}_{(K_1+K_2+K_3) \times 2} = \begin{bmatrix} -\mathbf{x}_{jt} & 0 \\ -\mathbf{v}_{jt} & 0 \\ \frac{\partial \xi_{jt}}{\partial \alpha} & \frac{\partial \omega_{jt}}{\partial \alpha} \\ \frac{\partial \xi_{jt}}{\partial \theta_2} & \frac{\partial \omega_{jt}}{\partial \theta_2} \\ 0 & -\mathbf{x}_{jt} \\ 0 & -\mathbf{w}_{jt} \end{bmatrix}, \quad \Omega_t \equiv \underbrace{\begin{bmatrix} \sigma_{\xi_t}^2 & \sigma_{\xi_t \omega_t} \\ \sigma_{\xi_t \omega_t} & \sigma_{\omega_t}^2 \end{bmatrix}}_{2 \times 2}.$$

I replace co-linear elements with zeros using  $\odot \Theta$

$$(G_{jt}\Omega_t^{-1}) \odot \Theta = \frac{1}{\sigma_\xi^2 \sigma_\omega^2 - \sigma_{\xi\omega}^2} \cdot \begin{bmatrix} -\sigma_\omega^2 x_{jt} & 0 \\ -\sigma_\omega^2 v_{jt} & \sigma_{\xi\omega} v_{jt} \\ \sigma_\omega^2 \frac{\partial \xi_{jt}}{\partial \alpha} - \sigma_{\xi\omega} \frac{\partial \omega_{jt}}{\partial \alpha} & \sigma_\xi^2 \frac{\partial \omega_{jt}}{\partial \alpha} - \sigma_{\xi\omega} \frac{\partial \xi_{jt}}{\partial \alpha} \\ \sigma_\omega^2 \frac{\partial \xi_{jt}}{\partial \theta_2} - \sigma_{\xi\omega} \frac{\partial \omega_{jt}}{\partial \theta_2} & \sigma_\xi^2 \frac{\partial \omega_{jt}}{\partial \theta_2} - \sigma_{\xi\omega} \frac{\partial \xi_{jt}}{\partial \theta_2} \\ 0 & -\sigma_\xi^2 x_{jt} \\ \sigma_{\xi\omega} w_{jt} & -\sigma_\xi^2 w_{jt} \end{bmatrix}.$$

Now we can partition our instrument set by column into “demand” and “supply”:

$$Z_{jt}^{opt,D} \equiv \underbrace{\mathbb{E}[(G_{jt}(Z_t)\Omega_t^{-1} \odot \Theta)_{\cdot 1} | \chi_t]}_{K_1 + K_2 + (K_3 - K_x)}, \quad Z_{jt}^{opt,S} \equiv \underbrace{\mathbb{E}[(G_{jt}(Z_t)\Omega_t^{-1} \odot \Theta)_{\cdot 2} | \chi_t]}_{K_2 + K_3 + (K_1 - K_x)}.$$

## Aside: What does Supply tell us about Demand?

**Demand**

$$\partial \alpha : \sigma_{\omega}^2 \frac{\partial \xi_{jt}}{\partial \alpha} - \sigma_{\xi \omega} \frac{\partial \omega_{jt}}{\partial \alpha}$$

$$\partial \tilde{\theta}_2 : \sigma_{\omega}^2 \frac{\partial \xi_{jt}}{\partial \tilde{\theta}_2} - \sigma_{\xi \omega} \frac{\partial \omega_{jt}}{\partial \tilde{\theta}_2}$$

**Supply**

$$\sigma_{\xi}^2 \frac{\partial \omega_{jt}}{\partial \alpha} - \sigma_{\xi \omega} \frac{\partial \xi_{jt}}{\partial \alpha}$$

$$\sigma_{\xi}^2 \frac{\partial \omega_{jt}}{\partial \tilde{\theta}_2} - \sigma_{\xi \omega} \frac{\partial \xi_{jt}}{\partial \tilde{\theta}_2}$$

- ▶ These are **cross equation restrictions**
- ▶ They serve as **overidentifying restrictions** for  $\theta_2$  parameters.
- ▶ This is the what imposing supply side tells us about demand (and *vice versa*)

How to construct optimal instruments in form of Chamberlain (1987). Start with initial instruments

$$\chi_t = A(\mathbf{X}_t, \mathbf{W}_t, \mathbf{V}_t)$$

$$\mathbb{E} \left[ \frac{\partial \xi_{jt}}{\partial \theta} | \chi_t \right] = \left[ \beta, \mathbb{E} \left[ \frac{\partial \xi_{jt}}{\partial \alpha} | \chi_t \right], \mathbb{E} \left[ \frac{\partial \xi_{jt}}{\partial \tilde{\theta}_2} | \chi_t \right] \right]$$

Some challenges:

1.  $p_{jt}$  or  $\eta_{jt}$  depends on  $(\omega_j, \xi_t)$  in a highly nonlinear way (no explicit solution!).
2.  $\mathbb{E} \left[ \frac{\partial \xi_{jt}}{\partial \tilde{\theta}_2} \mid X_t, w_t \right] = \mathbb{E} \left[ \left[ \frac{\partial \boldsymbol{\sigma}_t}{\partial \boldsymbol{\delta}_t} \right]^{-1} \left[ \frac{\partial \boldsymbol{\sigma}_t}{\partial \tilde{\theta}_2} \right] \mid Z_{jt}^D \right]$  (not conditioned on endogenous  $p$ !)

Things are **infeasible** because we don't know  $\theta_0$ !



## Feasible Recipe (BLP 1999)

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1. Fix  $\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3)$  and draw  $(\xi^*, \omega^*)$  from empirical density
2. Solve firm FOC's for  $\hat{\mathbf{p}}_t(\xi^*, \omega^*, \hat{\theta})$
3. Solve shares  $\sigma_t(\hat{\mathbf{p}}_t, \hat{\theta})$
4. Compute necessary Jacobian
5. Average over multiple values of  $(\xi^*, \omega^*)$ . (Lazy approach: use only  $(\xi^*, \omega^*) = 0$ ).

In simulation the “lazy” approach does just as well. (At least for iid normal  $(\xi, \omega)$ )

Alternative: Can we use  $\mathbb{E}[\mathbf{p}_t \mid \mathbf{Z}_t]$  instead for (2) if we don't have supply side

- Optimal instruments are easier to work out if  $p = mc$ .

$$c = p + \underbrace{\Delta^{-1}\sigma}_{\rightarrow 0} = X\gamma_1 + W\gamma_2 + \omega$$

- Linear cost function means linear reduced-form price function (could do nonlinear regression too)

$$\begin{aligned}\mathbb{E}\left[\frac{\partial \xi_{jt}}{\partial \alpha} | z_t\right] &= \mathbb{E}[p_{jt} | z_t] = x_{jt}\gamma_1 + w_{jt}\gamma_2 \\ \mathbb{E}\left[\frac{\partial \omega_{jt}}{\partial \alpha} | z_t\right] &= 0, \quad \mathbb{E}\left[\frac{\partial \omega_{jt}}{\partial \tilde{\theta}_2} | z_t\right] = 0 \\ \mathbb{E}\left[\frac{\partial \xi_{jt}}{\partial \tilde{\theta}_2} | z_t\right] &= \mathbb{E}\left[\frac{\partial \delta_{jt}}{\partial \tilde{\theta}_2} | z_t\right]\end{aligned}$$

- If we are worried about endogenous oligopoly markups is this a reasonable idea?
- Turns out that the important piece tends to be **shape** of jacobian for  $\Sigma$ .
- In either case what we care about is  $\mathbb{E}[p \mid x, z]$  (the **first stage**). Nothing is free here !

Table 2: Bias and Efficiency with Imperfect Competition

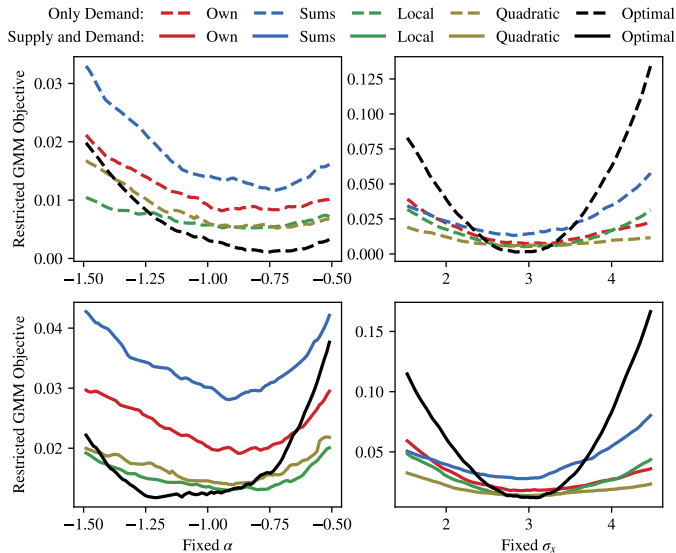
| Single Equation GMM |      |            |        |       |            |        |       |            |        |       |
|---------------------|------|------------|--------|-------|------------|--------|-------|------------|--------|-------|
|                     |      | $g_{jt}^1$ |        |       | $g_{jt}^2$ |        |       | $g_{jt}^3$ |        |       |
|                     | True | Bias       | St Err | RMSE  | Bias       | St Err | RMSE  | Bias       | St Err | RMSE  |
| $\beta^0$           | 2    | -0.127     | 0.899  | 0.907 | -0.155     | 0.799  | 0.814 | -0.070     | 0.514  | 0.519 |
| $\beta^1$           | 2    | -0.068     | 0.899  | 0.901 | 0.089      | 0.766  | 0.770 | -0.001     | 0.398  | 0.398 |
| $\alpha$            | -2   | 0.006      | 0.052  | 0.052 | 0.010      | 0.049  | 0.050 | 0.010      | 0.043  | 0.044 |
| $\sigma^1$          | 1    | -0.162     | 0.634  | 0.654 | -0.147     | 0.537  | 0.556 | -0.016     | 0.229  | 0.229 |
| Joint Equation GMM  |      |            |        |       |            |        |       |            |        |       |
|                     |      | $g_{jt}^1$ |        |       | $g_{jt}^2$ |        |       | $g_{jt}^3$ |        |       |
|                     | True | Bias       | St Err | RMSE  | Bias       | St Err | RMSE  | Bias       | St Err | RMSE  |
| $\beta^0$           | 2    | -0.095     | 0.714  | 0.720 | -0.103     | 0.677  | 0.685 | 0.005      | 0.459  | 0.459 |
| $\beta^1$           | 2    | 0.089      | 0.669  | 0.675 | 0.098      | 0.621  | 0.628 | -0.009     | 0.312  | 0.312 |
| $\alpha$            | -2   | 0.001      | 0.047  | 0.047 | 0.002      | 0.046  | 0.046 | -0.001     | 0.043  | 0.043 |
| $\sigma^1$          | 1    | -0.116     | 0.462  | 0.476 | -0.110     | 0.418  | 0.432 | 0.003      | 0.133  | 0.133 |

Bias, standard errors (St Err) and root mean squared errors (RMSE) are computed from 1000 Monte Carlo replications. Estimates are based on the MPEC algorithm and Sparse Grid integration. The instruments  $g_{jt}^1$ ,  $g_{jt}^2$  and  $g_{jt}^3$  are defined in section 2.4 and 2.5.

## IV Comparison: Conlon and Gortmaker (2020)

| Simulation | Supply | Instruments | Seconds | True Value |            |            |        | Median Bias |            |            |        | Median Absolute Error |            |            |        |
|------------|--------|-------------|---------|------------|------------|------------|--------|-------------|------------|------------|--------|-----------------------|------------|------------|--------|
|            |        |             |         | $\alpha$   | $\sigma_x$ | $\sigma_p$ | $\rho$ | $\alpha$    | $\sigma_x$ | $\sigma_p$ | $\rho$ | $\alpha$              | $\sigma_x$ | $\sigma_p$ | $\rho$ |
| Simple     | No     | Own         | 0.6     | -1         | 3          |            |        | 0.126       | -0.045     |            |        | 0.238                 | 0.257      |            |        |
| Simple     | No     | Sums        | 0.6     | -1         | 3          |            |        | 0.224       | -0.076     |            |        | 0.257                 | 0.208      |            |        |
| Simple     | No     | Local       | 0.6     | -1         | 3          |            |        | 0.181       | -0.056     |            |        | 0.242                 | 0.235      |            |        |
| Simple     | No     | Quadratic   | 0.6     | -1         | 3          |            |        | 0.206       | -0.085     |            |        | 0.263                 | 0.239      |            |        |
| Simple     | No     | Optimal     | 0.8     | -1         | 3          |            |        | 0.218       | -0.049     |            |        | 0.250                 | 0.174      |            |        |
| Simple     | Yes    | Own         | 1.4     | -1         | 3          |            |        | 0.021       | 0.006      |            |        | 0.226                 | 0.250      |            |        |
| Simple     | Yes    | Sums        | 1.5     | -1         | 3          |            |        | 0.054       | -0.020     |            |        | 0.193                 | 0.196      |            |        |
| Simple     | Yes    | Local       | 1.4     | -1         | 3          |            |        | 0.035       | -0.006     |            |        | 0.207                 | 0.229      |            |        |
| Simple     | Yes    | Quadratic   | 1.4     | -1         | 3          |            |        | 0.047       | -0.022     |            |        | 0.217                 | 0.237      |            |        |
| Simple     | Yes    | Optimal     | 2.2     | -1         | 3          |            |        | 0.005       | 0.012      |            |        | 0.170                 | 0.171      |            |        |
| Complex    | No     | Own         | 1.1     | -1         | 3          | 0.2        |        | -0.025      | 0.000      | -0.200     |        | 0.381                 | 0.272      | 0.200      |        |
| Complex    | No     | Sums        | 1.1     | -1         | 3          | 0.2        |        | 0.225       | -0.132     | -0.057     |        | 0.263                 | 0.217      | 0.200      |        |
| Complex    | No     | Local       | 1.0     | -1         | 3          | 0.2        |        | 0.184       | -0.107     | -0.085     |        | 0.274                 | 0.236      | 0.200      |        |
| Complex    | No     | Quadratic   | 1.0     | -1         | 3          | 0.2        |        | 0.200       | -0.117     | -0.198     |        | 0.299                 | 0.243      | 0.200      |        |
| Complex    | No     | Optimal     | 1.6     | -1         | 3          | 0.2        |        | 0.191       | -0.119     | 0.001      |        | 0.274                 | 0.195      | 0.200      |        |
| Complex    | Yes    | Own         | 3.9     | -1         | 3          | 0.2        |        | -0.213      | 0.060      | 0.208      |        | 0.325                 | 0.263      | 0.208      |        |
| Complex    | Yes    | Sums        | 3.3     | -1         | 3          | 0.2        |        | 0.018       | -0.104     | 0.052      |        | 0.203                 | 0.207      | 0.180      |        |
| Complex    | Yes    | Local       | 3.4     | -1         | 3          | 0.2        |        | -0.043      | -0.078     | 0.135      |        | 0.216                 | 0.225      | 0.200      |        |
| Complex    | Yes    | Quadratic   | 3.5     | -1         | 3          | 0.2        |        | -0.028      | -0.067     | 0.116      |        | 0.237                 | 0.227      | 0.200      |        |
| Complex    | Yes    | Optimal     | 4.9     | -1         | 3          | 0.2        |        | -0.024      | -0.036     | -0.002     |        | 0.193                 | 0.171      | 0.191      |        |

## IV Comparison: Conlon and Gortmaker (2020): Results



## Taking Stock

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30 years on, BLP mostly works

- ▶ There was some concern (see Knittel Metaxoglou 2014) that BLP estimation was a bit “fragile”
- ▶ PyBLP knows how to take derivatives, and we spent a lot of time tuning default options: solving for shares, integration rules, optimization, etc.
- ▶ Most problems people have today are: **too many parameters, not enough instruments** (or really bad ones).
- ▶ Cross-market variation in number of products, or characteristics of products would also help.

## The Bad News: Substitution

Your BLP model probably isn't as flexible as you think (Sorry!)

- ▶ The unrestricted matrix of  $D_{jk}(\mathbf{p})$ 's (or elasticities) is  $J \times J$  and probably not feasible to estimate directly. → need some **dimension reduction**.
- ▶ Logit simply assumes proportional substitution  $D_{jk} = \frac{\sigma_k}{1-\sigma_j}$  so that  $\mathbf{D}(\mathbf{p})$  is of rank one!
- ▶ The BLP solution is to project  $\mathbf{D}(\mathbf{p})$  onto a lower-dimensional basis of  $x_j$  characteristics.
  - Ultimately the basis will be only as good as the characteristics  $x_{jt}$  with heterogeneous coefficients.
  - Distributional assumptions on  $f(\beta_i)$  (ie: independent normal) further restrict the basis.
- ▶ The hardest thing to match is typically **substitution to closest substitutes**.
  - I worry that most BLP models (one RC, etc.) look too much like the plain logit.
- ▶ People often assume **HUGE outside good shares** and then have  $D_{j0} > 0.9$ .
  - We make fun of macro/trade for **monopolistic competition** but easy to estimate something close.



## The Bad News: Elasticities

If you aren't careful, can still get some weird economics...

- ▶ Plain logit assumes **elasticities increasing in  $p_j$**  and **markups that are decreasing**.
  - One RC or putting all products in one nest probably isn't going to fix that
- ▶ Simple versions tend to lead to **high income / price insensitive** consumers buying all the cars, yogurt, etc.
  - → Probably everyone should have a **correlated** RC on Price and Constant terms.
- ▶ How  $f(y_i - \alpha_i p_j)$  looks as it may determine **pass-through**  
Griffith, Nesheim O'Connell (2018), Birchall Verboven (2022), Miravete, Seim, Thurk (2023)
  - → Lognormal  $\alpha_i$  is probably a good idea
  - → bins for  $\alpha(y_i)$  by income also good (break distributional relationship between  $p_{jt}$  and  $y_{it}$ ).

## What we assumed away: Search and Consideration

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- ▶ Everything this week assumed **full information** and **full consideration**: every consumer evaluates every  $j \in \mathcal{J}_t$ .
- ▶ If consumers only consider  $C_i \subset \mathcal{J}$ , estimated elasticities conflate **preferences** with **attention**:
  - Goeree (2008): PCs — advertising and limited information.
  - Honka (2014): search and switching costs in auto insurance.
  - Honka, Hortaçsu, Vitorino (2017): advertising and awareness in banking.
  - Abaluck and Adams-Prassl (2021): recovering consideration from asymmetries in demand (menus).
- ▶ With search frictions, price responses mix “who searches” with “who prefers” — a different demand object.
- ▶ Search and consideration get several weeks of their own in ECON-GA 1802 (spring).

What does this mean:

- ▶ We should always check  $\mathbb{E}[p \mid x, z]$  before we do anything else.
- ▶ Can use FRAC to figure out where the heterogeneity is, get starting values
- ▶ May want to consider adding a supply side (if you're willing to assume for counterfactuals, why not?)
- ▶ Certainly should do `results.compute_optimal_instruments()` in PyBLP.